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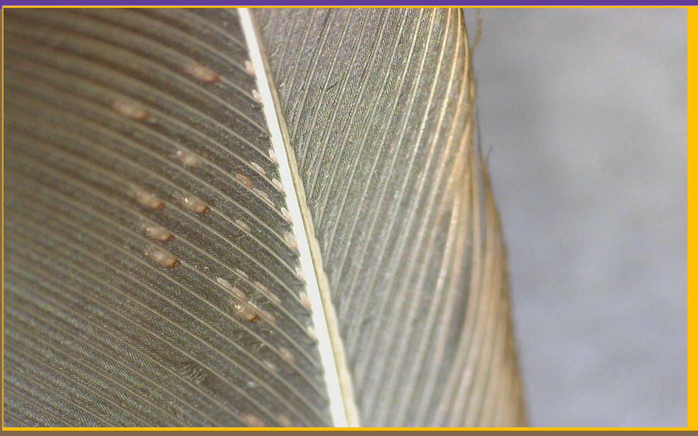

Mite Differentiation of Wisconsin Owls

Davis Lensert, Gaven Thompson, Lucy Hughes, Sam Coleman
University of Wisconsin –Stevens Point

Introduction

- Feather mites (subclass *Acari*) are considered parasitic with many species of birds, including owls (order *Strigiformes*).
- Some species of mites are known to cause diseases within owls, such as hyperkeratosis (Fig. 1).
- Feather mites are known to adapt similar physical characteristics to better suit their host-parasite relationship[1]. As such, many of them adopt similar proportional sizes to their host.
- Our prediction for this project follows:
  - H_0: No difference in proportional size in mite to owl parasite relationship
  - H_A: There is a difference in proportional size in mite to owl parasite relationship

Methods

- We made extensive use of Stephen J. Taft Animal Parasitology Collection as well as Ecdysis database to locate the mite specimens (Fig. 3) within the collection. We randomly selected 10 mites from each owl species, with an exception of the snowy owl, which had 9 total samples in the collection.
- We measured mite specimens using microscopes to determine length wise from the back of the abdomen to the frontmost leg, in micrometers (μm).
- We ran ANOVA: single factor (Fig. 2) to determine variation among mite to owl ratio sizes. We then ran Tukey's test (Table 1.) to determine differences among each owl species.

Results

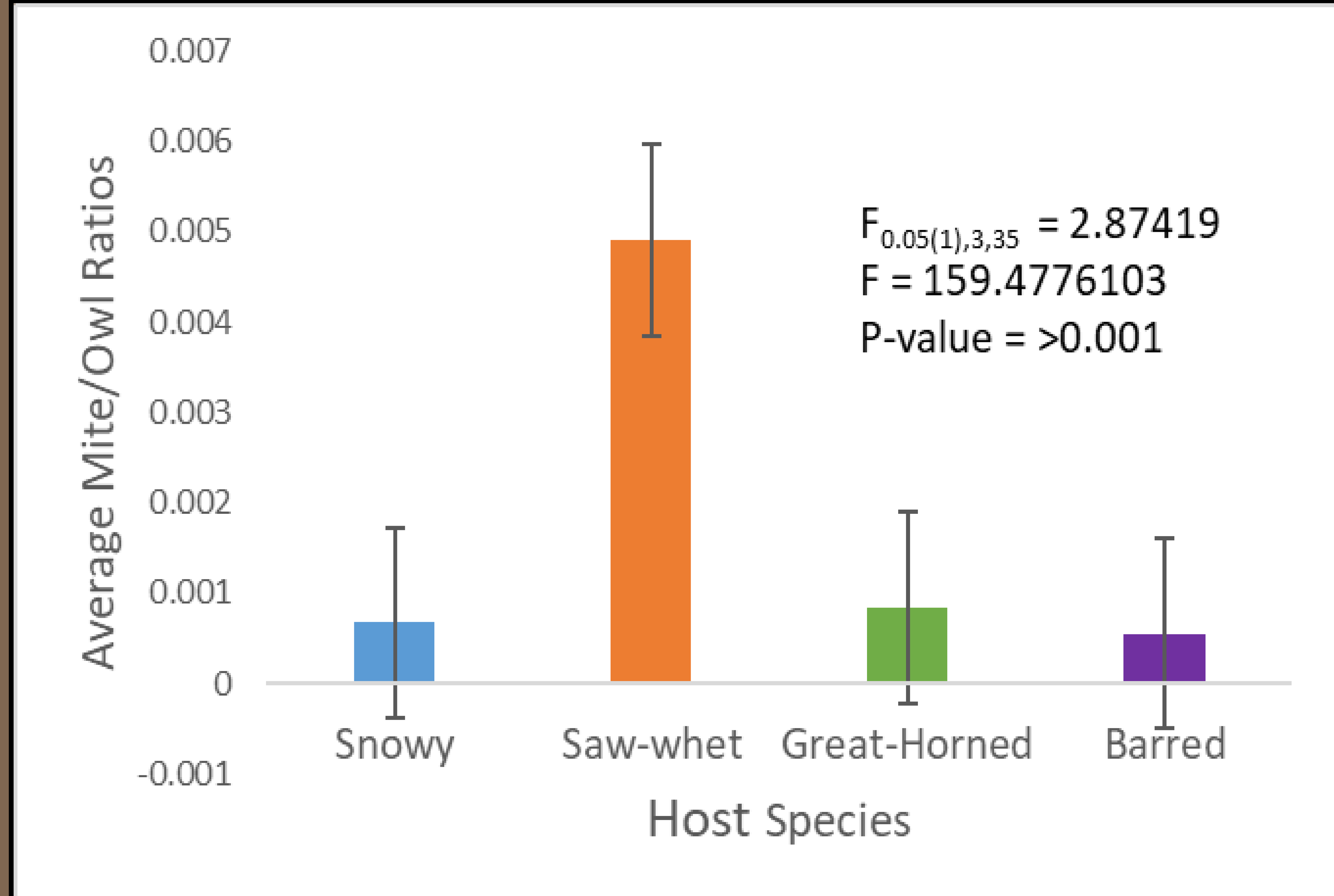

Fig 2. Mite to owl size ratios. We averaged the length (μm) of ten random mites per host species and divided it by the average length of each owl species After running ANOVA: single factor, we can reject our null hypothesis.

Table 1. Tukey's test can show us a pair-wise comparison of variation in the mite to owl size ratios. Saw-whet owls have significantly larger proportions than all other owls. Snowy, barred and great-horned owls' mite to owl size ratios are not significantly different from each other.

Host Species	AVG Mite to Owl Ratio	SE = sqrt(within MS/n)	n=sample size / group	
Saw-Whet	0.004907692	0.000168891	9.75	
Great-Horned	0.000831776			
Snowy	0.000666667			
Barred	0.000551515			
Comparison of Ratios in Ascending Order		q=(avgB-avgA)/SE	q(0.05)(35)(6)	Statement
Saw-Whet - Barred	0.004356177	25.79290034	3.823	Reject
Saw-Whet - Snowy	0.004241026	25.11108887	3.823	Reject
Saw-Whet - Great-Horned	0.004075917	24.13347921	3.823	Reject
Great-Horned - Barred	0.000280261	1.65942113	3.823	Fail to Reject
Great-Horned - Snowy	0.000165109	0.977609659	3.823	Fail to Reject
Snowy - Barred	0.000115152	0.681811471	3.823	Fail to Reject

Discussion

- A significant difference was observed.
- We encountered a couple of hurdles when investigating the mite specimens, one such being the inability to identify specific species of mites.
- Further studies could investigate why mites on saw-whet owls were so much larger than those on other species.
- Studies could also add larger sample sizes for both owl and mite specimens, and in particular use multiple owl specimens for mite collection.

Fig 1. Hyperkeratosis in a Great-horned owl[2]

Fig 3. Feather mite found on Barred Owl

References

[1] Mironov, S. V. "Morphological adaptations of feather mites to different types of plumage and skin of birds." Паразитологический сборник Зоологического института Академии наук СССР (том 34) (1987): 131.

[2]Terry A. Schulz et al. Knemidokoptes mutans (*Acari: Knemidocoptidae*) in a Great-horned Owl (*Bubo virginianus*). J Wildl Dis 1 July 1989; 25 (3): 430–432. doi: https://doi.org/10.7589/0090-3558-25.3.430